

(04) Representing Floating-Point

Task One

Convert the number 0.001011 to floating-point representation.

There are 16 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

Sign bit (1 bit)	Mantissa (15 bits)	Exponent (8 bits)
0	101 1000 0000 0000	1111 1110

Task Two

Convert the number 100.0011 to floating-point representation.

There are 8 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

Sign bit (1 bit)	Mantissa (15 bits)	Exponent (8 bits)
0	100 0011 0000 0000	0000 0011

Task Three

Convert the number 0.00001011 to floating-point representation.

There are 16 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

Sign bit (1 bit)	Mantissa (15 bits)	Exponent (8 bits)
0	101 1000 0000 0000	1111 1100

Task Four

Convert the number -10.0101 to floating-point representation.

There are 16 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

Sign bit (1 bit)	Mantissa (15 bits)	Exponent (8 bits)
1	100 1010 0000 0000	0000 0010

Task Five

Convert the number -0.010101 to floating-point representation.

There are 16 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

Sign bit (1 bit)	Mantissa (15 bits)	Exponent (8 bits)
1	101 0100 0000 0000	1111 1111